LAB ASSIGNMENT-4

TEAM MEMBERS:

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The task is to implement and visualize the fundamental concepts of the two-dimensional Discrete Fourier Transform (2-D DFT).

The objectives include generating the 8×8 2-D DFT basis functions, creating a binary image containing a rectangular region, computing the 2-D DFT of the image using the direct summation formula (without using any built-in FFT/DFT functions), and analyzing the effect of frequency centering.

The given task is addressed using a single MATLAB script, which sequentially performs basis generation, image creation, manual DFT computation, and spectrum visualization:

How to Run the Program

1. Compile the matlab script file and run the script.
2. Then, enter the required inputs:
    - Top-left row index of the rectangle
    - Top-left column index of the rectangle
    - Width of the rectangle
    - Height of the rectangle
3. The corresponding results are displayed.

2D_DFT.m

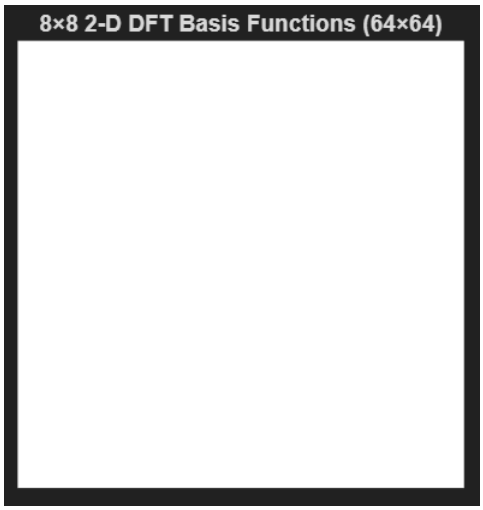
```
clc;
clear;
close all;
% step1: generating the basis of an 8×8 2-D DFT and display it as a 64×64 image
N = 8;
basis_img = zeros(N*N, N*N);
for u = 0:N-1
   for v = 0:N-1
       basis = zeros(N, N);
       for x = 0:N-1
           for y = 0:N-1
               basis(x+1, y+1) = exp(-1j*2*pi*(u*x/N + v*y/N));
           end
       end
       % placing basis magnitude into 64x64 image
       row = u*N + 1;
       col = v*N + 1;
       basis_img(row:row+N-1, col:col+N-1) = abs(basis);
   end
end
figure;
imshow(mat2gray(basis_img));
title('8×8 2-D DFT Basis Functions (64×64)');
% step2: creating a binary 64×64 image containing a rectangle
M = 64;
N = 64;
img = zeros(M, N);
% user inputs
x0 = input('Enter top-left row index (0-63): ');
y0 = input('Enter top-left column index (0-63): ');
w  = input('Enter rectangle width: ');
h  = input('Enter rectangle height: ');
% creating rectangle
img(x0+1:x0+h, y0+1:y0+w) = 1;
figure;
imshow(img);
title('Binary Rectangle Image');
% step3: Compute and plot the 2-D DFT for the image from step2
DFT = zeros(M, N);
for u = 0:M-1
   for v = 0:N-1
       sum_val = 0;
       for x = 0:M-1
           for y = 0:N-1
               sum_val = sum_val + img(x+1,y+1) * ...
                   exp(-1j*2*pi*(u*x/M + v*y/N));
           end
```

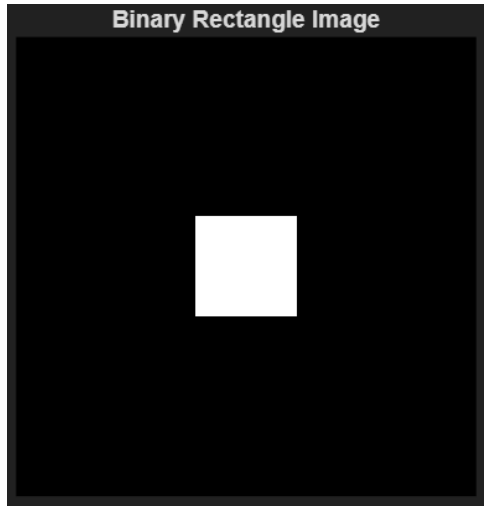
```
        end
        DFT(u+1,v+1) = sum_val;
    end
end
figure;
imshow(log(1 + abs(DFT)), []);
title('Magnitude Spectrum (Uncentered DFT)');
% step4: Compute and plot the 2-D DFT for the centered image
centered_img = zeros(M,N);
for x = 0:M-1
    for y = 0:N-1
        centered_img(x+1,y+1) = img(x+1,y+1) * (-1)^(x+y);
    end
end
DFT_centered = zeros(M, N);
for u = 0:M-1
    for v = 0:N-1
        sum_val = 0;
        for x = 0:M-1
            for y = 0:N-1
                sum_val = sum_val + centered_img(x+1,y+1) * ...
                    exp(-1j*2*pi*(u*x/M + v*y/N));
            end
        end
        DFT_centered(u+1,v+1) = sum_val;
    end
end
figure;
imshow(log(1 + abs(DFT_centered)), []);
title('Magnitude Spectrum (Centered DFT)');
```

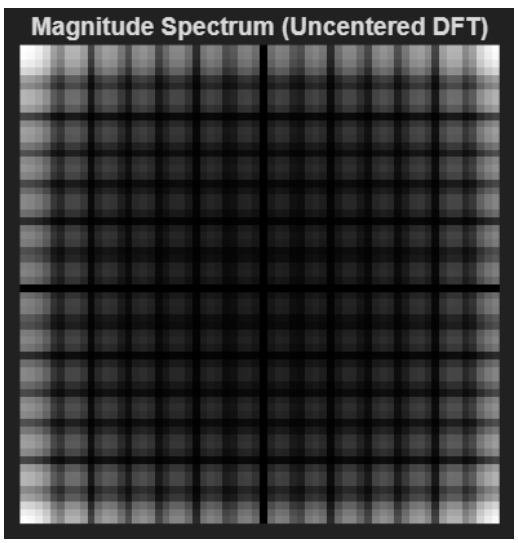
Results:

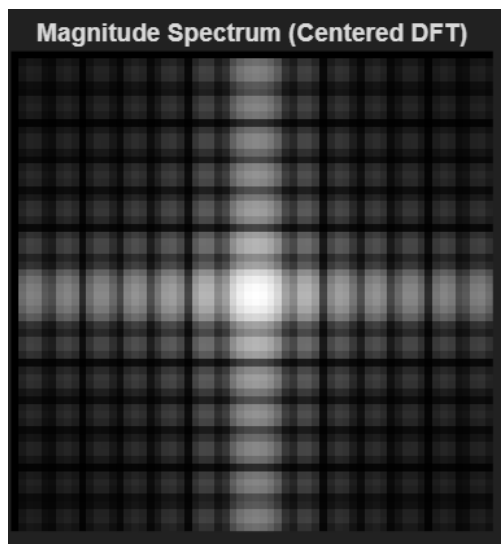
1. For centered Rectangle(symmetric):

```
Enter top-left row index (0-63): 25
Enter top-left column index (0-63): 25
Enter rectangle width: 14
Enter rectangle height: 14
```



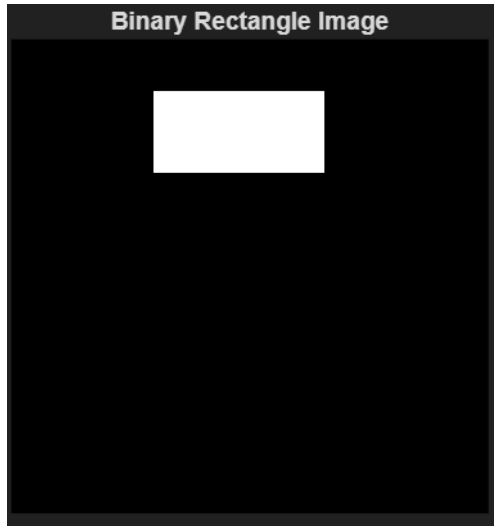


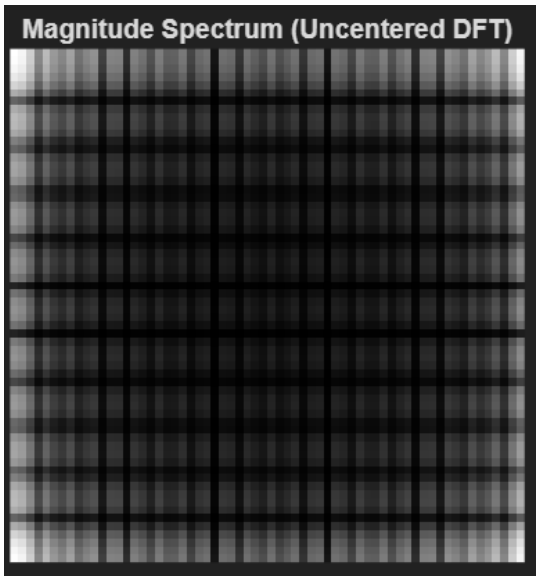


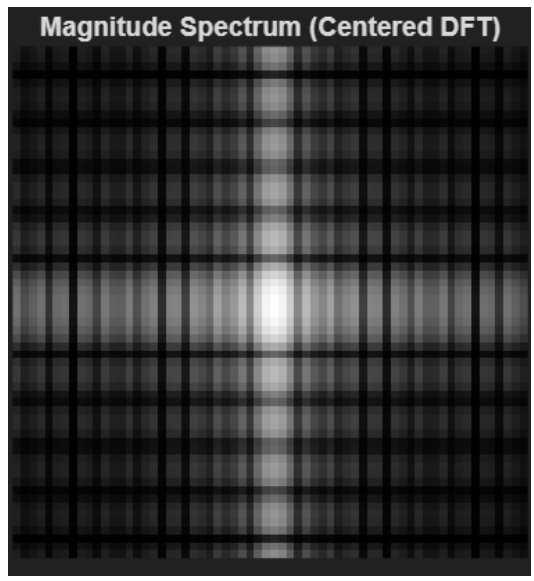


2. For Non-Centered Rectangle (asymmetric):

```
Enter top-left row index (0-63): 7
Enter top-left column index (0-63): 19
Enter rectangle width: 23
Enter rectangle height: 11
```

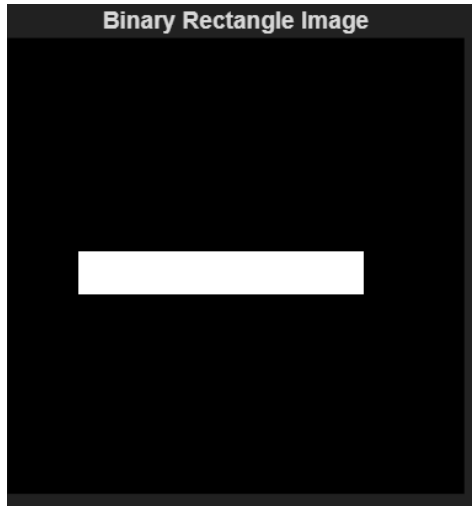


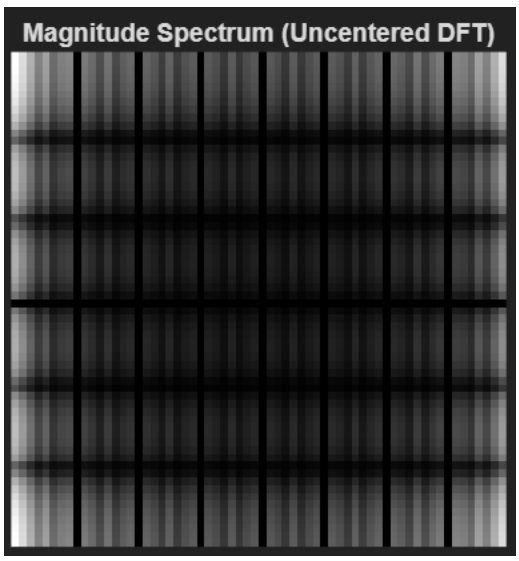


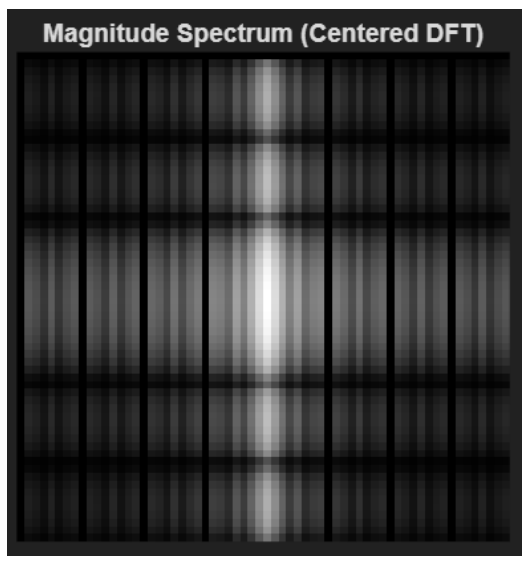


3. Horizontal Rectangle:

```
Enter top-left row index (0-63): 30
Enter top-left column index (0-63): 10
Enter rectangle width: 40
Enter rectangle height: 6
```

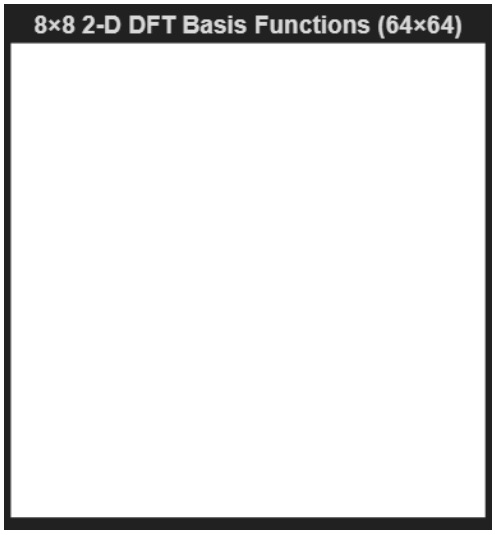


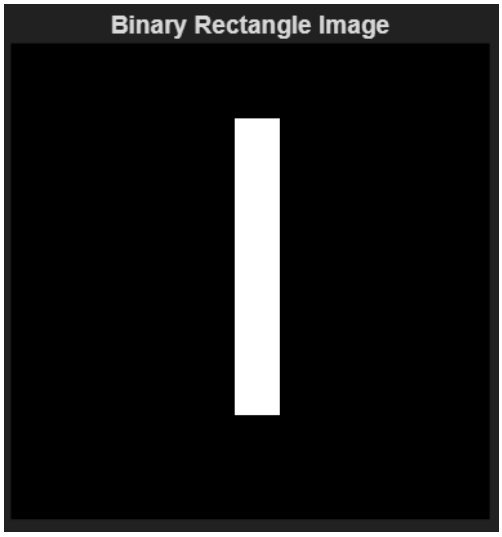


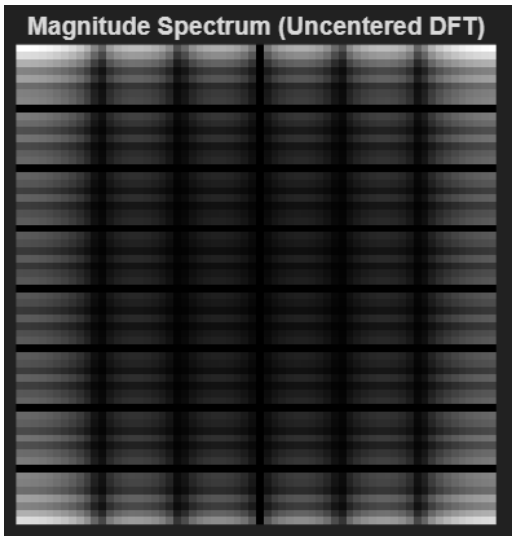


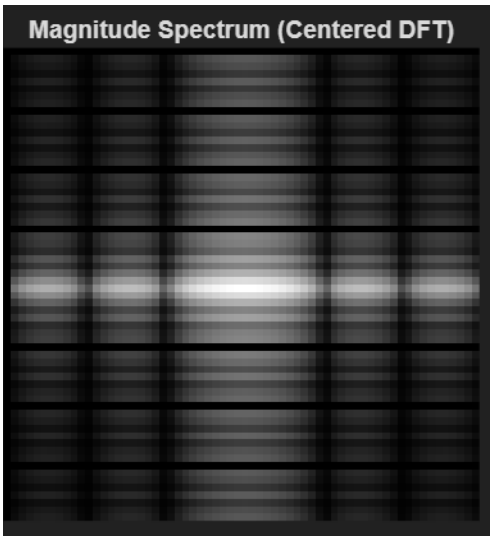
4. Vertical Rectangle:

```
Enter top-left row index (0-63): 10
Enter top-left column index (0-63): 30
Enter rectangle width: 6
Enter rectangle height: 40
```


8×8 2-D DFT Basis Functions (64×64)


Binary Rectangle Image


Magnitude Spectrum (Uncentered DFT)


Magnitude Spectrum (Centered DFT)

Extension: Implementation of 2-D FFT for Performance Comparison

In the initial implementation, the two-dimensional Discrete Fourier Transform was computed using direct summation in accordance with the problem constraints. As an extension to the assignment, a Fast Fourier Transform (FFT)-based implementation is introduced to compare computational efficiency and spectral equivalence between manual DFT and FFT-based approaches.

How the Extension betters

1. FFT produces the same magnitude spectrum as manual DFT
2. FFT is much faster
3. FFT requires explicit centering (fftshift)

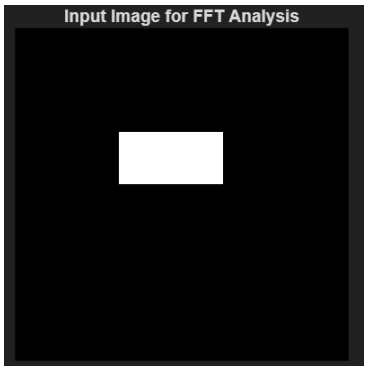
FFT_Extension.m

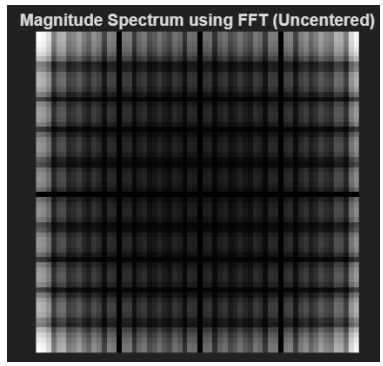
```
clc;
clear;
close all;
% step1: generating image
M = 64;
N = 64;
img = zeros(M, N);
% user inputs
x0 = input('Enter top-left row index (0-63): ');
y0 = input('Enter top-left column index (0-63): ');
w  = input('Enter rectangle width: ');
h  = input('Enter rectangle height: ');
theta_deg = input('Enter rotation angle (degrees): ');
theta = theta_deg * pi / 180;
% rectangle center
xc = x0 + h/2;
yc = y0 + w/2;
for x = 0:M-1
   for y = 0:N-1
       xp = x + 0.5;
       yp = y + 0.5;
       % inverse rotation
       xr =  (xp - xc)*cos(theta) + (yp - yc)*sin(theta);
       yr = -(xp - xc)*sin(theta) + (yp - yc)*cos(theta);
       if abs(xr) <= h/2 && abs(yr) <= w/2
           img(x+1, y+1) = 1;
       end
   end
end
figure;
imshow(img, []);
title('Input Image for FFT Analysis');
% step2: uncentered 2D fft
FFT_img = fft2(img);
figure;
imshow(log(1 + abs(FFT_img)), []);
title('Magnitude Spectrum using FFT (Uncentered)');
% step3: centered fft
FFT_centered = fftshift(FFT_img);
figure;
imshow(log(1 + abs(FFT_centered)), []);
title('Magnitude Spectrum using FFT (Centered)');
```

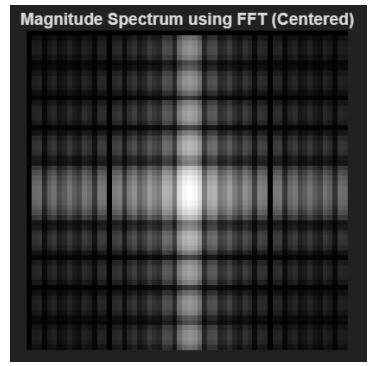
Results:

1. Axis-Aligned Rectangle (Baseline FFT):

```
Enter top-left row index (0-63): 20
Enter top-left column index (0-63): 20
Enter rectangle width: 20
Enter rectangle height: 10
Enter rotation angle (degrees): 0
```

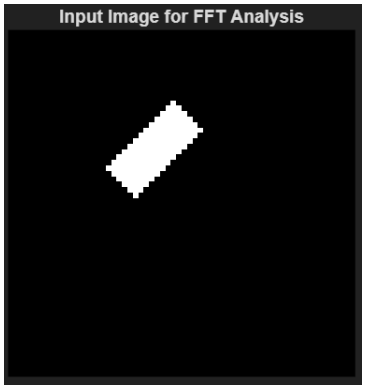


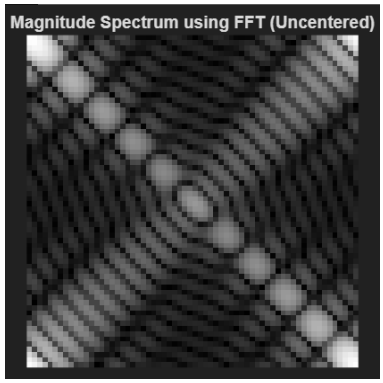


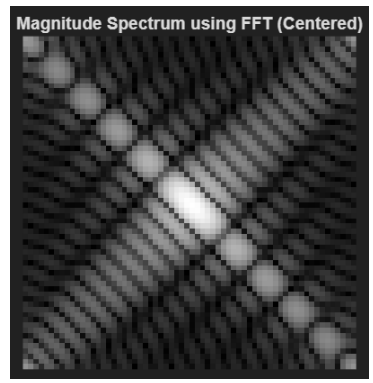


2. Rotated Rectangle (Diagonal Spectrum):

```
Enter top-left row index (0-63): 18
Enter top-left column index (0-63): 18
Enter rectangle width: 18
Enter rectangle height: 8
Enter rotation angle (degrees): 45
>>
```

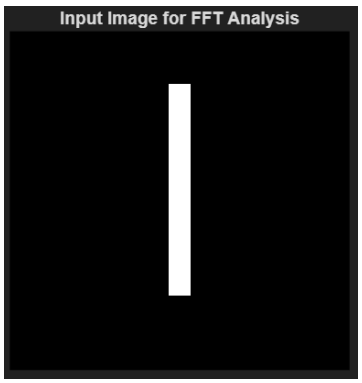


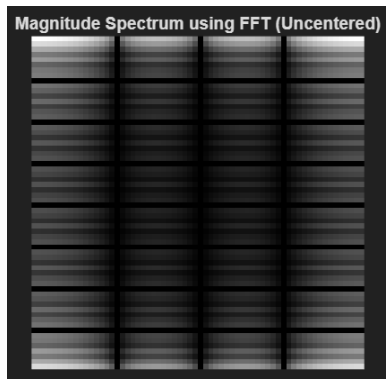


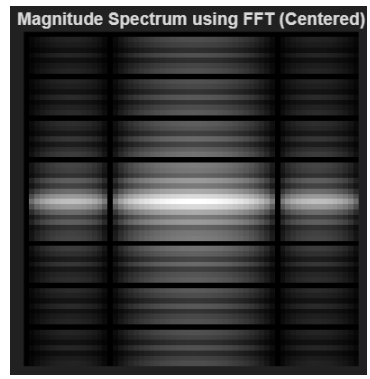


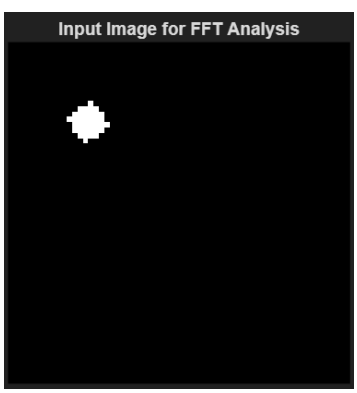
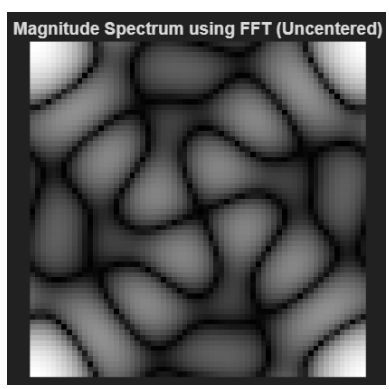
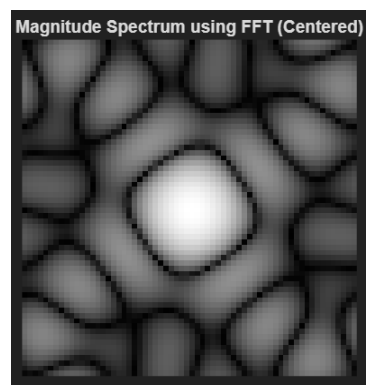
3. Thin Vertical Rectangle:

```
Enter top-left row index (0-63): 10
Enter top-left column index (0-63): 30
Enter rectangle width: 4
Enter rectangle height: 40
Enter rotation angle (degrees): 0
```







4. Small Rotated Rectangle (High Frequency Content)

```
Enter top-left row index (0-63): 12
Enter top-left column index (0-63): 12
Enter rectangle width: 6
Enter rectangle height: 6
Enter rotation angle (degrees): 30
```